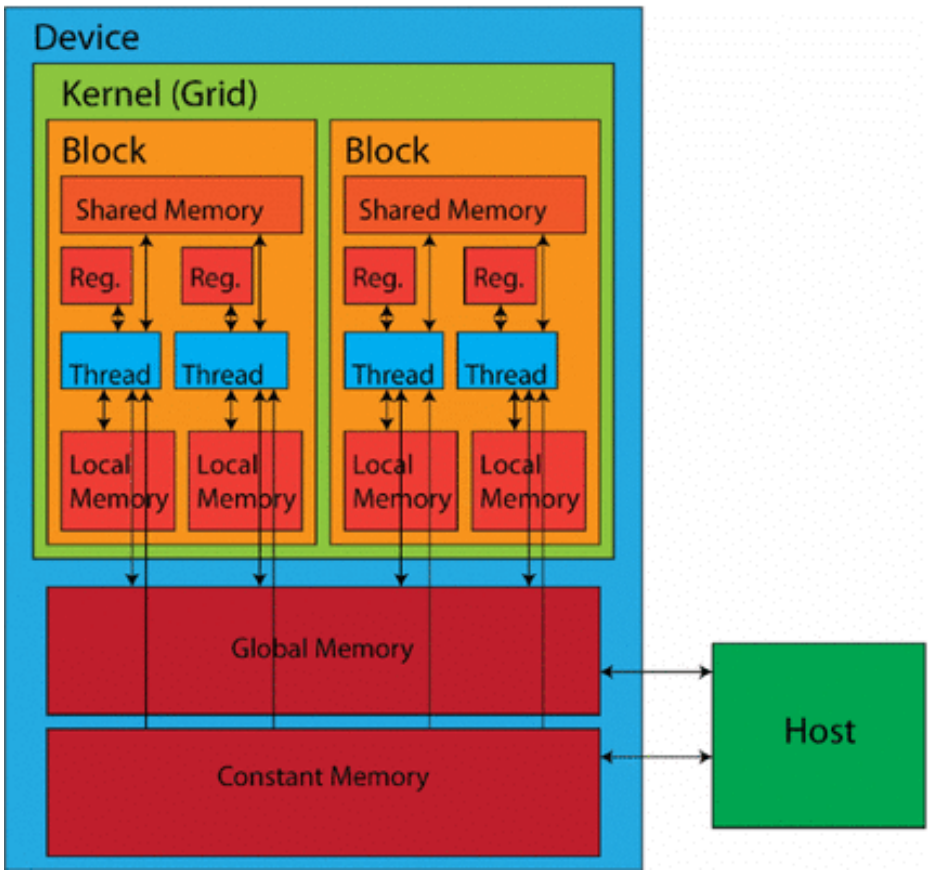
GPU Architecture:

Introduction to GPU Architecture

Graphics Processing Units (GPUs) are specialized processors designed to accelerate graphics rendering and parallel computations. Modern GPUs have evolved into highly parallel, multithreaded, manycore processors with tremendous computational horsepower and very high memory bandwidth



Key Architectural Features

1. Parallel Processing Design

- **Massive parallelism**: GPUs contain thousands of smaller, more efficient cores (compared to CPUs)
- **SIMT architecture**: Single Instruction, Multiple Thread execution model
- **Deeply pipelined**: Designed to hide memory latency through massive multithreading

2. Core Components

- **Streaming Multiprocessors (SMs)**: Fundamental processing units containing:
    - CUDA cores (NVIDIA) or Stream processors (AMD)
    - Special function units (SFUs)
    - Tensor cores (in modern GPUs for AI acceleration)
- **Memory hierarchy**:
    - Global memory (high-capacity GDDR/HBM)
    - Shared memory/L1 cache
    - Registers (fastest, per-thread)
- **Interconnect**: High-speed bus between components

3. Memory Architecture

- **Hierarchical memory model** with different access speeds
- **Coalesced memory access** patterns for optimal performance
- **Memory controllers** optimized for high bandwidth

Comparison with CPU Architecture

Feature	CPU	GPU
Cores	Few (2-64)	Many (1000s)
Threads per core	1-2 (typically)	100s
Cache hierarchy	Large, complex	Smaller, simpler
Best for	Serial code	Parallel workloads
Clock speed	Higher	Lower